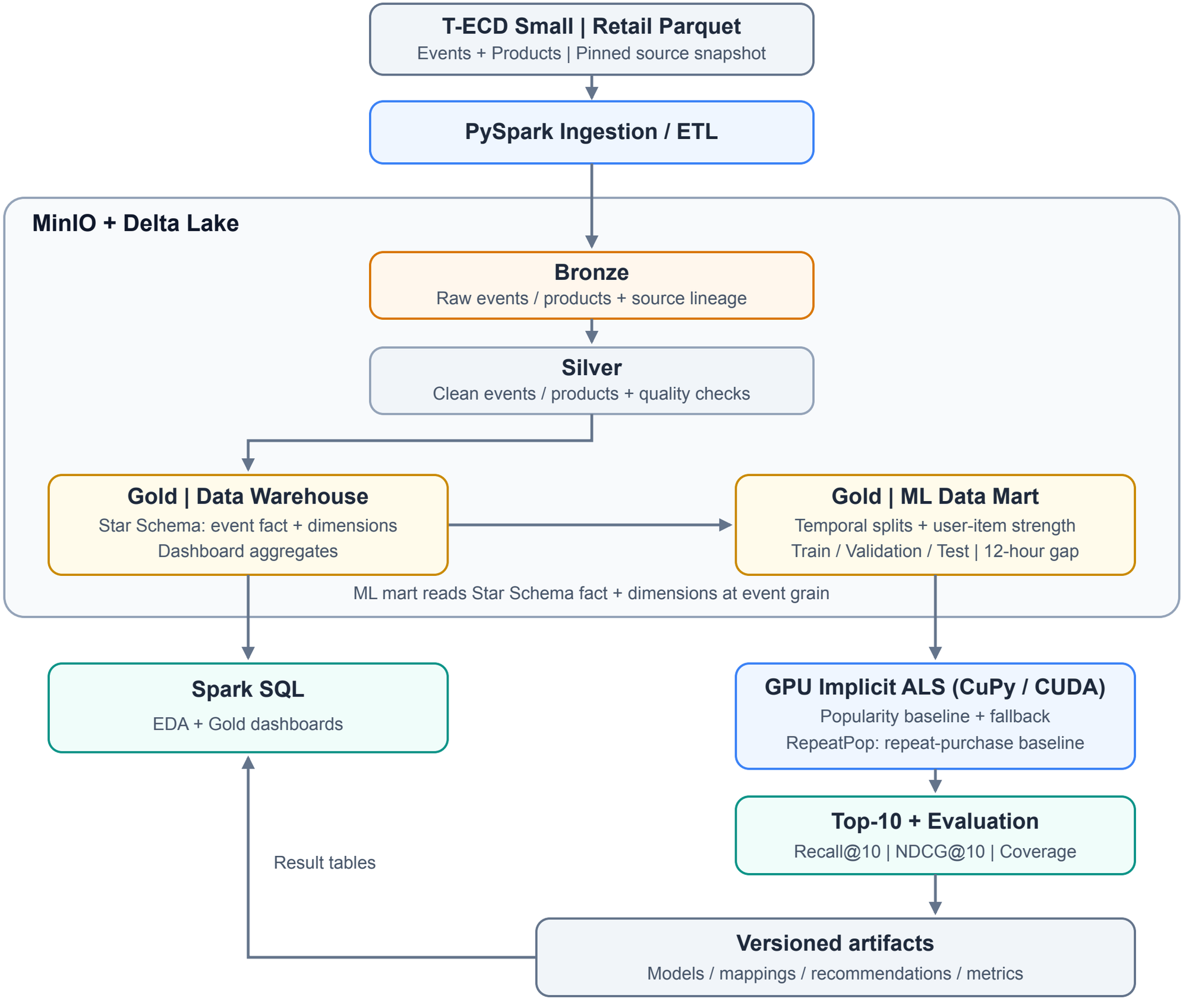
T-ECD Small | Retail Parquet
Events + Products | Pinned source snapshot
PySpark Ingestion / ETL
MinIO + Delta Lake
Bronze
Raw events / products + source lineage
Silver
Clean events / products + quality checks
Gold | Data Warehouse
Star Schema: event fact + dimensions
Dashboard aggregates
Gold | ML Data Mart
Temporal splits + user-item strength
Train / Validation / Test | 12-hour gap
ML mart reads Star Schema fact + dimensions at event grain
Spark SQL
EDA + Gold dashboards
GPU Implicit ALS (CuPy / CUDA)
Popularity baseline + fallback
RepeatPop: repeat-purchase baseline
Top-10 + Evaluation
Recall@10 | NDCG@10 | Coverage
Result tables
Versioned artifacts
Models / mappings / recommendations / metrics
Approved design | PySpark ETL and SQL | GPU training | RAM, SSD spill, bounded swap fallback